Breast Tissue Biopsies 11

11.1 Overview

A major emphasis in the Ghana Breast Health Study will be on classifying tumors by hormone receptor and growth factor receptor status. This study will allow investigators, in an African population, to assess distinctive risk factor profiles for tumor subtypes. In order to get an accurate picture of the prevalence of molecular subtypes of tumors, we will attempt to collect tissue samples at biopsy from Cases enrolled in the study. If a Case had a breast biopsy prior to her enrollment in the study (either at KBTH, KATH, PLH, or elsewhere), study staff should attempt to obtain tissue from that biopsy. If the original diagnostic core biopsy cannot be obtained, a tissue sample should be obtained from the excisional surgery if the Case undergoes breast surgery while in the study.

11.2 Breast Biopsy Procedures

Since the majority of women present with palpable lumps, Cook Medical, Quick-Core® 16-gauge Biopsy Needles will be used for diagnosis. Typically, when a needle core biopsy is used for diagnosis, the surgeon collects 4-6 tissue cores for women with a suspicious breast lump for pathologic assessment and diagnosis. To assure quality tissue samples for molecular marker assessment, each hospital (KBTH, KATH, and PLH) will be provided with labeled pre-filled formalin containers for immediate fixation. This will enable the samples to be immediately fixed, preventing problems in assessing tumor tissue markers.

The protocol for collection of breast biopsies is designed to optimize histopathologic preservation and maintain biomolecular integrity in a setting where tissue processing cannot occur within the normal 24 (+/-8hr) hour window on a vacuum-enabled tissue processor. The protocol has been validated in the National Cancer Institute, Tissue Array Research Program (TARP) Lab, and is currently employed by the Genotype-Tissue Expression study.

11.2.1 Breast Tissue Core Fixation Procedure

The pathologist will use the following breast tissue core fixation procedure for the study.

1. Using a 16-gauge biopsy needle, the surgeon will obtain 4-6 breast tissue cores.

2. One to two cores will placed in each labeled cassette.

3. Up to three cassettes with 1-2 tissue cores will be placed in one prefilled 60ml container with 30ml 10% neutral buffered formalin.

4. The surgeon should complete the Breast Tissue Biopsy Form (Appendix 11-1) at the time the breast tissue biopsy cores are obtained. Instructions for completing the form are in Appendix 11-2.

5. Fixation of the tissue cores should proceed at room temperature (not to exceed 40° C) for 16-32 hours.

6. The tissue cassette should be tapped to remove excess fixative, and wiped with a tissue.

7. After 16-32 hours in formalin, the core biopsy specimens will be processed by hospital pathologists to make a diagnosis using their current tissue processing and embedding procedures.

8. Any extra cores not needed for diagnosis will be labeled and placed in a separate cassette and shipped to the NIH for processing. . The cassettes should be labelled with the Accession number followed by a hyphen and block number (-1, -2, -3). A log of samples with Accession number, date of collection, and corresponding PID should be maintained.

9. The cassette(s) are then transferred to a Seal-A-Meal® bag, and vacuum sealed with a Seal-A-Meal® vacuum unit.

10. The sealed vacuum bag can be stored for up to six weeks at room temperature (not to exceed 40° C).

11. Biopsy samples (fixed cores or paraffin blocks) will be periodically shipped at room temperature to the NCI Biorepository, where they will be stored and used for molecular pathology, including inclusion in the construction of tissue microarrays. Vacuum sealed bags should be packed in a Styrofoam shipping container provided by NCI's shipping contractor, appropriately marked for contents according to the manifest agreed upon with NCI.

12. Up to 12 cassettes can be packed and shipped per bag. (Refer to Chapter 12, Shipping Biospecimens, for specifications on shipping the biopsy samples.)

NOTES:

- Fixative should be maintained at room temperature (not to exceed 40° C) prior to use. Fixative with precipitate or past the expiration date should not be used. Do not store fixative containers in direct light.
- No residual fixative is present during shipping, and does not need to be noted on shipping documents as a chemical hazard. Specimens should be shipped as "diagnostic specimens/tissue" rather than biologic specimens, as they are chemically preserved.
- If excisional biopsies are performed, the same procedures noted above should be followed. The pathologist should core the tissue from the excisional biopsy and then follow steps 1-11.

11.2.2 Tissue Processing On Receipt at AMPL

Vacuum sacks are opened, inventoried and cassettes are placed in 70% ETOH for a minimum 24 hours prior to processing.

11.2.3 Tissue Processing Protocol at AMPL

The AMPL pathologist will use the following tissue processing procedure for the study.

1. First, the water from the tissues must be removed by dehydration. This is usually done by immersing tissues in a series of alcohols (e.g., 70% to 95% to 100%).
2. The next step is called "clearing" and consists of removal of the dehydrant with a substance that will be miscible with the embedding medium (paraffin). The most common clearing agent is xylene.
3. Finally, the tissue is infiltrated with paraffin.
    a. Paraffins can be purchased that differ in melting point and in various consistencies. A product called paraplast contains added plasticizers that make the paraffin blocks easier for some technicians to cut.

A vacuum can be applied inside the tissue processor to assist penetration of the embedding agent. The above processes are almost always automated for the large volumes of routine tissues processed. Automation consists of an instrument that moves the tissues through the various agents on a preset time scale. Newer processors are controlled by computers rather than cam wheels and have sealed reagent wells to which a vacuum and/or heat can be applied.

Table 11-1. TARP Standard Processing Times Using Tissue Tek VIP 5

Solution	Conc (%)	Time (hr:min)	Temp (Celsius)	P/V	Mix
Formalin	10	—	—	ON	SLOW
Formalin	10	—	—	ON	SLOW
Alcohol	70	30 min	—	ON	SLOW
Alcohol	95	40 min	—	ON	SLOW
Alcohol	95	40 min	—	ON	SLOW
Alcohol	100	40 min	—	ON	SLOW
Alcohol	100	40 min	—	ON	SLOW
Alcohol	100	40 min	—	ON	SLOW
Xylene	—	45 min	—	ON	SLOW
Xylene	—	45 min	—	ON	SLOW
Paraffin	—	30 min	60	ON	SLOW
Paraffin	—	30 min	60	ON	SLOW
Paraffin	—	30 min	60	ON	SLOW
Paraffin	—	30 min	60	ON	SLOW

Processing times can be altered according to preference and programs can be modified for scheduled finish time and any alterations to processing.

Tissues that come off the tissue processor are still in the cassettes and must be manually put into the blocks by a technician who must pick the tissues out of the cassette and pour molten paraffin over them. This "embedding" process is very important, because the tissues must be aligned, or oriented, properly in the block of paraffin.

(See Embedding Protocol for materials and procedure)

11.2.4 Maintenance

The following maintenance procedures must be followed for the study.

1. After each run, the processor must operate a cleaning cycle to remove excess paraffin in the chamber with washes of xylene and 100% alcohol.

2. Reagents should be changed every five cycles or as needed (depending on the volume of specimens processed).